

RF HiRes Dash — User Manual

Complete Guide for RF Engineers

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Quick Start (5 Minutes)

Get started with RF HiRes Dash in 5 minutes.

1. Click "  TX Site" → place transmitter on map
2. Click "  Add RX" → place multiple receiver sites
3. Adjust RF parameters: frequency, ERP, heights, conductivity
4. Click "  Run PtP Audit" → see results
5. Click "  Export" → choose DOCX or KMZ
6. Click "  Log Out" → session auto-saves!

RF Parameters Explained

Frequency (MHz): 300 kHz – 2 GHz valid range. Examples: 100 (FM), 800 (mobile), 2400 (WiFi).

ERP (dBm): Effective Radiated Power = Power (dBm) + Antenna Gain (dBi) – Loss (dB).

System Loss (dB): Feedline and connector losses. Typical: 0–5 dB.

TX Height (m): Antenna height above ground. Examples: Ground 5–30m, Rooftop 20–50m, Tower 50–300m.

RX Height (m): Receiver antenna height. Examples: Hand-held 1.5m, Mobile 1.5–2.5m, Fixed 5–20m.

Ground Conductivity (S/m): Most important parameter after distance/frequency.

- Typical values:
 - Sea water: 5.0 S/m
 - Wet soil: 0.01–0.1 S/m
 - Dry land: 0.005 S/m
 - Dry sand: 0.0001 S/m

Propagation Models & Frequency Coverage

Medium Wave (300 kHz – 30 MHz)

For MW frequencies, RF HiRes Dash uses ITU-R P.368-9 standard ground wave model.

Results validated against official ITU-R P.368-9 propagation tables.

Covers AM broadcast and HF radio.

VHF/UHF (30 MHz – 2 GHz)

Uses Fresnel-Kirchhoff diffraction combined with two-ray ground reflection.

Covers FM broadcast, mobile cellular, WiFi, UHF TV.

Above 2 GHz (Experimental)

Results not validated. Use at your discretion for 5G and millimeter-wave.

Session Management & Saving Work

Auto-Save: Logout and Save

Click "  Log Out" at top right.

Your session automatically saves: TX location, RX sites, RF parameters, results.

Next login: everything restores as you left it.

Session Export/Import (JSON)

Export: Click "  Export" → "  Save Session (JSON)" to download session.json file.

Import: Click "  Import Session" and select your saved JSON file.

Use cases: Share analysis, archive projects, restore backups, compare scenarios.

Region Defaults

Click "  Load Region" dropdown in sidebar.

Select your region or terrain type (Coastal Australia, Urban, Alpine, etc.).

App sets default conductivity, heights, thresholds for that region.

You can adjust all parameters manually afterward.

Batch RX Sites Import

EASIEST WAY: Export from Map and Edit

Build your RX list directly on the map, then export and reuse:

1. Click "📍 Add RX" → place receiver sites on map
2. Right-click each RX marker → edit name ("Sydney CBD", "North Tower", etc.)
3. Click "📄 Export" → "Export CSV" to download your RX list
4. Open CSV in Excel → save as XLSX format (add H_RX column if needed)
5. Click "📁 Import RX List" → select XLSX file → all sites reload with custom names

RX List Format (XLSX)

Required columns:

- RX-ID: Latitude,longitude (e.g., "-33.8688,151.2093")
- RX Name: Site identifier (e.g., "Sydney CBD")
- H_RX (m): Receiver antenna height (optional, defaults 3m)

Example:

- -33.8688,151.2093 | Sydney CBD | 2.5
- -33.9425,151.1912 | Parramatta | 3.0
- -34.0195,151.0850 | Penrith | 2.0

Batch TX Database Import

Import multiple transmitter sites via XLSX file.

TX Database Columns

Required (11 columns):

- tx_name, lat, lon, freq_mhz, erp_w, ant_type, sys_loss_db, h_tx_m, sigma_sm, sigma_tx_sm, h_rx_m
- Optional: thresholds, coverage radius, cell size, opacity, network name, organization

How to Import

6. 1. Click "  Import TX Database" in sidebar
7. 2. Select your XLSX file
8. 3. App validates file and shows any errors
9. 4. TX sites load into dropdown selector
10. 5. Select a TX → parameters load into sidebar
11. 6. Run "  Run PtP Audit" or "  Compute Coverage Grid"
12. 7. Export results as DOCX or KMZ

Exporting Results

DOCX Format

Professional report: RF parameters, propagation model, results table, map with heatmap, elevation profile.

KMZ Format

Google Earth overlay: TX marker (red), RX markers (blue), color-coded heatmap.

CSV Format

Results table for spreadsheet analysis.

Troubleshooting

- Login: Check spam folder, wait 30 seconds, try again.
- Slow compute: Reduce search radius or grid cell size.
- Export disabled: Run at least one propagation computation first.
- Map issues: Refresh page (Ctrl+R) or log out and back in.